

$$\begin{aligned}
 75. (\sqrt{12} + \sqrt{18}) - (\sqrt{3} + \sqrt{2}) &= (\sqrt{4 \times 3} + \sqrt{9 \times 2}) - (\sqrt{3} + \sqrt{2}) \\
 &= (2\sqrt{3} + 3\sqrt{2}) - (\sqrt{3} + \sqrt{2}) \\
 &= (2\sqrt{3} - \sqrt{3}) + (3\sqrt{2} - \sqrt{2}) = \sqrt{3} + 2\sqrt{2}.
 \end{aligned}$$

$$76. \frac{\sqrt{24} + \sqrt{216}}{\sqrt{96}} = \frac{\sqrt{4 \times 6} + \sqrt{36 \times 6}}{\sqrt{16 \times 6}} = \frac{2\sqrt{6} + 6\sqrt{6}}{4\sqrt{6}} = \frac{8\sqrt{6}}{4\sqrt{6}} = 2.$$

$$\begin{aligned}
 77. \frac{\sqrt{80} - \sqrt{112}}{\sqrt{45} - \sqrt{63}} &= \frac{\sqrt{16 \times 5} - \sqrt{16 \times 7}}{\sqrt{9 \times 5} - \sqrt{9 \times 7}} \\
 &= \frac{4\sqrt{5} - 4\sqrt{7}}{3\sqrt{5} - 3\sqrt{7}} = \frac{4(\sqrt{5} - \sqrt{7})}{3(\sqrt{5} - \sqrt{7})} = \frac{4}{3} = 1\frac{1}{3}.
 \end{aligned}$$

$$\begin{aligned}
 78. 3\sqrt{5} + \sqrt{125} &= 17.88 \Rightarrow 3\sqrt{5} + \sqrt{25 \times 5} = 17.88 \\
 \Rightarrow 3\sqrt{5} + 5\sqrt{5} &= 17.88 \Rightarrow 8\sqrt{5} = 17.88 \Rightarrow \sqrt{5} = 2.235. \\
 \therefore \sqrt{80} + 6\sqrt{5} &= \sqrt{16 \times 5} + 6\sqrt{5} = 4\sqrt{5} + 6\sqrt{5} \\
 &= 10\sqrt{5} = (10 \times 2.235) = 22.35.
 \end{aligned}$$

$$79. \sqrt{50} \times \sqrt{98} = \sqrt{50 \times 98} = \sqrt{4900} = 70.$$

$$\begin{aligned}
 80. \text{Given exp.} &= \sqrt{4 \times 2} + 2\sqrt{16 \times 2} - 3\sqrt{64 \times 2} + 4\sqrt{25 \times 2} \\
 &= 2\sqrt{2} + 8\sqrt{2} - 24\sqrt{2} + 20\sqrt{2} = 6\sqrt{2} = 6 \times 1.414 = 8.484.
 \end{aligned}$$

$$\begin{aligned}
 81. \text{Given exp.} &= \frac{3\sqrt{12}}{2\sqrt{28}} \times \frac{\sqrt{98}}{2\sqrt{21}} = \frac{3\sqrt{4 \times 3}}{2\sqrt{4 \times 7}} \times \frac{\sqrt{49 \times 2}}{2\sqrt{21}} \\
 &= \frac{6\sqrt{3}}{4\sqrt{7}} \times \frac{7\sqrt{2}}{2\sqrt{21}} = \frac{21\sqrt{6}}{4\sqrt{7 \times 21}} = \frac{21\sqrt{6}}{28\sqrt{3}} \\
 &= \frac{3}{4}\sqrt{2} = \frac{3}{4} \times 1.414 = 3 \times 0.3535 = 1.0605.
 \end{aligned}$$

$$\begin{aligned}
 82. \text{Given exp.} &= \sqrt{\frac{11025}{100}} \times \sqrt{\frac{1}{100}} \div \sqrt{\frac{25}{10000}} - \sqrt{\frac{42025}{100}} \\
 &= \frac{105}{10} \times \frac{1}{10} \div \frac{5}{100} - \frac{205}{10} \\
 &= \frac{105}{100} \times \frac{100}{5} - \frac{205}{10} \\
 &= 21 - \frac{205}{10} = \frac{5}{10} = \frac{1}{2} = 0.50.
 \end{aligned}$$

83. Sum of decimal places in the numerator and denominator under the radical sign being the same, we remove the decimal.

$$\therefore \text{Given exp.} = \sqrt{\frac{81 \times 484}{64 \times 625}} = \frac{9 \times 22}{8 \times 25} = 0.99.$$

$$84. \text{Given exp.} = \sqrt{\frac{204 \times 42}{7 \times 34}} = \sqrt{36} = 6.$$

$$\begin{aligned}
 85. \text{Given exp.} &= \sqrt{\frac{81 \times 324 \times 4624}{15625 \times 289 \times 729 \times 64}} \\
 &= \frac{9 \times 18 \times 68}{125 \times 17 \times 27 \times 8} = \frac{3}{125} = 0.024.
 \end{aligned}$$

$$\begin{aligned}
 86. \text{Given exp.} &= \sqrt{\frac{9.5 \times .08500}{.19 \times .0017}} = \sqrt{\frac{95 \times 8500}{19 \times 17}} \\
 &= \sqrt{5 \times 500} = \sqrt{2500} = 50.
 \end{aligned}$$

$$\begin{aligned}
 87. \text{Given exp.} &= \sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{\left(\frac{0.03}{10}\right)^2 + \left(\frac{0.21}{10}\right)^2 + \left(\frac{0.065}{10}\right)^2}} \\
 &= \sqrt{\frac{100[(0.03)^2 + (0.21)^2 + (0.065)^2]}{(0.03)^2 + (0.21)^2 + (0.065)^2}} = \sqrt{100} = 10.
 \end{aligned}$$

$$\begin{aligned}
 88. \sqrt{(7 + 3\sqrt{5})(7 - 3\sqrt{5})} &= \sqrt{(7)^2 - (3\sqrt{5})^2} \\
 &= \sqrt{49 - 45} = \sqrt{4} = 2.
 \end{aligned}$$

$$\begin{aligned}
 89. \left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)^2 &= (\sqrt{3})^2 + \left(\frac{1}{\sqrt{3}}\right)^2 - 2 \times \sqrt{3} \times \frac{1}{\sqrt{3}} \\
 &= 3 + \frac{1}{3} - 2 = 1 + \frac{1}{3} = \frac{4}{3}.
 \end{aligned}$$

$$\begin{aligned}
 90. \sqrt{4a^2 - 4a + 1 + 3a} \\
 &= \sqrt{(1)^2 + (2a)^2 - 2 \times 1 \times 2a + 3a} = \sqrt{(1 - 2a)^2 + 3a} \\
 &= (1 - 2a) + 3a = (1 + a) = (1 + 0.1039) = 1.1039.
 \end{aligned}$$

$$\begin{aligned}
 91. \sqrt{\frac{(0.75)^3}{(1 - 0.75)} + [0.75 + (0.75)^2 + 1]} \\
 &= \sqrt{\frac{(0.75)^3 + (1 - 0.75)[(1)^2 + (0.75)^2 + 1 \times 0.75]}{1 - 0.75}} \\
 &= \sqrt{\frac{(0.75)^3 + [(1)^3 - (0.75)^3]}{1 - 0.75}} \\
 &= \sqrt{\frac{1}{0.25}} = \sqrt{\frac{100}{25}} = \sqrt{4} = 2.
 \end{aligned}$$

$$\begin{aligned}
 92. a + b + c &= 27\sqrt{29} \Rightarrow 2c + \frac{3}{2}c + c = 27\sqrt{29} \\
 \Rightarrow \frac{9}{2}c &= 27\sqrt{29} \Rightarrow c = 6\sqrt{29}.
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{a^2 + b^2 + c^2} &= \sqrt{(a + b + c)^2 - 2(ab + bc + ca)} \\
 &= \sqrt{(27\sqrt{29})^2 - 2\left(2c \times \frac{3}{2}c + \frac{3}{2}c \times c + c \times 2c\right)} \\
 &= \sqrt{(729 \times 29) - 2\left(3c^2 + \frac{3}{2}c^2 + 2c^2\right)} \\
 &= \sqrt{(729 \times 29) - 2 \times \frac{13}{2}c^2} \\
 &= \sqrt{(729 \times 29) - 13 \times (6\sqrt{29})^2} = \sqrt{29(729 - 468)} \\
 &= \sqrt{29 \times 261} = \sqrt{29 \times 29 \times 9} = 29 \times 3 = 87.
 \end{aligned}$$

$$93. \sqrt{0.\overline{4}} = \sqrt{\frac{4}{9}} = \frac{2}{3} = 0.666..... = 0.\overline{6}.$$

$$94. \sqrt{0.09} = \sqrt{\frac{9}{100}} = \frac{3}{10} = 0.3, \text{ which is rational.}$$